FSMamba: A Frequency-Enhanced Spatial–Temporal Mamba Network for Local–Global Remote Sensing Image Change Detection

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***Abstract*—Remote sensing change detection (RSCD) is challenged by complex scene variations, illumination differences, misregistration errors, and the need to jointly model global semantics and fine-grained local structures. To address these issues, this article proposes a frequency-enhanced spatiotemporal Mamba (FSMamba) network for local–global RSCD. FSMamba integrates frequency-domain analysis, spatiotemporal interaction, and efficient long-range dependence modeling within a unified framework. Specifically, a frequency-domain enhancement module (FDEM) is designed to capture global structural discrepancies via learnable spectral filtering, effectively suppressing pseudo-changes caused by illumination variations and noise. A Spatio-Temporal perception module (STPM) leverages channelwise interaction and the linear-complexity visual state-space model (VSSM) to enhance semantic consistency between bitemporal features. In addition, a global–local collaborative decoder (GLCD) is introduced to adaptively fuse multiscale features, preserving large-scale structural integrity while maintaining fine-grained boundary details. Extensive experiments on five public benchmark datasets demonstrate that FSMamba consistently outperforms state-of-the-art convolutional neural network (CNN), Transformer, and Mamba-based methods in terms of accuracy, robustness, and computational efficiency, validating its effectiveness under complex RSCD scenarios. Our source code is available on https://github.com/2419924948gyt/FSMamba**

***Index Terms*—Frequency-domain analysis, Mamba, remote sensing change detection (RSCD), state-space model (SSM).**

I. Introduction

REMOTE sensing change detection (RSCD) is a fundamental task in Earth observation data analysis, aiming to identify and quantify surface changes from multitemporal observations. By characterizing the spatiotemporal evolution of geographical objects and environmental factors, RSCD plays a critical role in numerous applications, including urban expansion analysis [1], [2], forest and ecosystem degradation monitoring [3], [4], land-use and land-cover dynamics investigation [5], [6], and natural disaster assessment and postevent recovery monitoring [7], [8].

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With the increasing availability of very-high-resolution (VHR) remote sensing imagery [9], [10], fine-grained change analysis has been significantly enhanced. In addition, hyperspectral image (HSI) has attracted growing interest due to its rich spectral information, and comprehensive surveys [11] have summarized its challenges and methodological advances. However, this progress also introduces substantial challenges [12]. Radiometric inconsistencies caused by illumination variations, seasonal differences, and sensor viewing geometry, together with object misregistration in complex scenes, often produce pseudo-changes unrelated to actual surface evolution. Effectively suppressing such pseudo-changes while fully exploiting rich spatial details has, therefore, become a critical issue for accurate and robust RSCD.

Early RSCD approaches mainly relied on pixel-level spectral information, such as image differencing [13] and change vector analysis [14], which directly measure pixelwise differences between multitemporal images. These methods are highly sensitive to radiometric variations and sensor noise, and performance depends on empirically selected thresholds. Subsequent studies introduced handcrafted shallow features, including principal component analysis (PCA)-based statistical features [15] and object-based strategies [16] that incorporate spatial structure. Although detection stability is partially improved, these methods remain vulnerable to pseudo-changes in complex VHR scenes, thereby suffering from poor generalization performance and high false-alarm rates.

With the rapid development of deep learning [17], convolutional neural network (CNN)-based methods have substantially advanced RSCD performance through powerful feature representation and end-to-end learning. Daudt et al. [18] first introduced fully convolutional networks (FCNs) into RSCD, proposing fully convolutional early fusion (FC-EF) and two Siamese variants, FC-Siam-Conc and FC-Siam-Diff, which established a foundational paradigm. Fang et al. [19] further proposed a densely connected Siamese network to enhance multiscale feature fusion and spatiotemporal modeling between bitemporal images. Building upon these frameworks, several representative approaches such as image fusion network (IFNet) [20] and UNet-based variants [21] have been developed by exploiting the task-specific characteristics of RSCD. In heterogeneous remote sensing scenarios, graph-based contrastive learning has also been explored to mitigate modality discrepancies [22], where super pixel-level relational modeling is employed to enhance cross-sensor change dis-

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crimination. To overcome the inherent locality of convolution, enlarged receptive fields [23] and attention mechanisms [24] have been introduced to enhance contextual modeling capability. Despite these efforts, convolution operations remain limited in capturing long-range dependencies and global contextual information [25], and CNN-based models still struggle to fully characterize complex spatial structures and high-level semantic relationships in VHR imagery.

The emergence of visual Transformer architectures [26], [27], [28] enables explicit modeling of global dependencies through self-attention mechanisms. Various Transformer-based RSCD methods, such as bitemporal Transformer (BIT) [29], ChangeFormer [30], and BiFA [31], have reported improved performance in boundary delineation, semantic consistency, and robustness to pseudo-changes. However, the quadratic scaling of computational complexity with image patch length imposes a significant computational burden, rendering Transformer-based models inefficient for dense prediction tasks, including binary change detection. To improve computational efficiency, some methods adopt sparse attention mechanisms [32], [33], [34] or adjust the attention window size [27], [35]. However, these designs are achieved at the expense of a reduced global receptive field.

Recently, state-space models (SSMs) [36], [37] have enabled efficient global dependence modeling through selective state-space mechanisms and dynamic state updates, achieving linear computational complexity with respect to sequence length. Building upon this paradigm, methods based on Vision Mamba [38] and VMamba [39] have explored the application of SSMs to visual representation learning [40], [41], [42], [43] and RSCD tasks [44], [45]. Most existing Mamba-based methods directly operate on temporal and spatial dimensions, focusing on sequential modeling and spatial dependence learning, while largely overlooking explicit frequency-domain representations.

However, standard visual Mamba blocks inherently rely on spatial-domain sequential scanning, lacking explicit mechanisms to filter out high-frequency pseudo-changes commonly observed in RSCD, such as misregistration-induced boundary artifacts. Because the selective state-space mechanism continuously updates its latent states along spatial trajectories, localized high-frequency perturbations are easily absorbed and propagated across the sequence, leading to fragmented false positives (FPs). In contrast, genuine structural discrepancies and nonsemantic high-frequency noise are highly separable in the frequency domain. Motivated by this observation, we hypothesize that introducing an explicit frequency-domain prior can help mitigate this vulnerability. Specifically, we design a frequency-domain enhancement module (FDEM) to decouple and suppress high-frequency disturbances. To avoid disrupting Mamba's continuous state dynamics, FDEM operates in parallel with the Spatio-Temporal perception module (STPM). The critical connection between these complementary paradigms is realized at the representation level within a global–local collaborative decoder (GLCD), where frequency-enhanced priors and selectively modeled spatiotemporal contexts are adaptively fused.

To address these challenges and instantiate the aforementioned hypothesis, this article proposes a novel frequency-enhanced spatiotemporal Mamba (FSMamba) network for RSCD. By unifying frequency-domain analysis and visual SSM (VSSM) within a parallel architecture, FSMamba effectively filters out high-frequency pseudo-changes while capturing dynamic long-range dependencies, thereby ensuring accurate change reconstruction in complex scenarios.

The main contributions of this work are summarized as follows.

1) Proposed FSMamba, a frequency-enhanced spatiotemporal framework for RSCD, which introduces an explicit frequency-domain complementary branch alongside VSSM to mitigate sensitivity to pseudo-changes.
2) Proposed an FDEM to capture global structural discrepancies via learnable spectral filtering, improving robustness to illumination variations and noise.
3) Proposed an STPM that leverages the linear-complexity property of Mamba-based models to capture long-range dependencies and reduce false alarms and missed detections.
4) Proposed a GLCD that enables adaptive multiscale feature fusion, improving boundary accuracy and the structural integrity of change objects.
5) Extensive experiments on five public benchmark datasets demonstrating consistent advantage over state-of-the-art CNN, Transformer, and Mamba-based methods in terms of accuracy, robustness, and generalization.

II. Related Works

A. CNN-Based RSCD Methods

During the early development stage of RSCD, CNNs were widely recognized as the dominant paradigm due to their robust capacity for local feature extraction and spatial representation learning. Daudt et al. [18] first introduced FC-EF and two variants (FC-Siam-Conc and FC-Siam-Diff), which are built upon FCN and a Siamese network. Zhang et al. [46] proposed a supervised Siamese network with an improved triplet loss for discriminative optical aerial image change detection. To enhance change discrimination and computational efficiency, Lei et al. [47] proposed a difference enhancement and spatial–spectral nonlocal network (DESSN), which integrates difference enhancement modeling with spatial–spectral nonlocal operations to improve object integrity and reduce redundant computation. In addition, attention mechanisms have been increasingly adopted in CNN-based RSCD models to reinforce contextual awareness. Zhang et al. [20] proposed a deeply supervised IFNet that leverages two-stream feature extraction and attention-based multilevel feature fusion for accurate change detection in high-resolution bitemporal remote sensing images. Chen and Shi [48] designed a spatiotemporal attention module to highlight salient change regions and suppress background noise. Raza et al. [49] proposed an efficient nested convolutional network (EUNet-CD) with attention-based multiscale fusion collectively. To address the demand for robust bitemporal change detection and promote large-scale aerial image change detection research, Shi et al. [50] proposed

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deeply supervised attention metric-based network (DSAMNet) via the combination of deeply supervised metric learning and attention mechanisms.

Despite these advances, CNN-based RSCD methods remain constrained by localized receptive fields and implicit spectral modeling, which limits global context capture and robustness to variations in illumination and imaging conditions. In this article, we propose a Mamba-based architecture that enables efficient global modeling and frequency-aware representation learning to suppress pseudo-changes and alleviate the aforementioned limitations.

B. Transformer-Based RSCD Methods

Transformers [26], [28] have introduced a paradigm shift in computer vision and RSCD tasks by leveraging self-attention to effectively capture long-range dependencies and global semantic relations. Anders et al. [29] first adopted a BIT encoder to jointly model cross-temporal correlations. Zhang et al. [51] presented SwinSUNet, which adopts a Siamese U-shaped architecture composed entirely of hierarchical Swin Transformer [27] blocks for multiscale global modeling. Building upon this idea, Bandara and Patel [30] presented ChangeFormer, a Siamese Transformer-based change detection framework that leverages hierarchical self-attention encoding and multilayer perceptron (MLP) decoding to model multiscale long-range context. Feng et al. [52] proposed a dual-branch multilevel intertemporal network (DMINet), which introduces a joint-attention mechanism that unifies self-attention and cross-attention to enable early intertemporal interaction and multilevel difference aggregation. In addition, Dong et al. [35] developed CSWin Transformer by combining cross-shaped window self-attention with locally enhanced positional encoding to improve modeling capacity under constrained computation. Overall, these studies have demonstrated the advantages of Transformers in modeling the global context and improving semantic consistency in complex scenes. Despite these advances, Transformer-based methods still face challenges in computational scalability and local-detail preservation. Hybrid CNN–Transformer designs have, therefore, been explored to alleviate these issues. Liu et al. [53] proposed a hybrid CNN–Transformer network with multiscale context aggregation network (MSCANet) to address intraclass variability and improve long-range context modeling in RSCD. Song et al. [54] introduced ACABFNet, which employs axial cross-attention to efficiently capture global dependencies and fuse local and global features in a CNN–Transformer bibranch architecture for change detection. Yang et al. [55] developed ConvFormer-CD, a CNN–Transformer hybrid model that employs temporal attention and a difference integration module to refine bitemporal change representations.

However, these hybrid approaches typically involve nontrivial tradeoffs between efficiency and accuracy. Moreover, most Transformer-based RSCD models perform feature differencing mainly in the spatial domain, overlooking the potential of frequency-domain representations for suppressing pseudo-changes and enhancing structural robustness. In this article, we propose an FSMamba framework that combines Fourier-domain enhancement with VSSM to enable linear-complexity global modeling and suppress pseudo-changes.

C. Mamba-Based RSCD Methods

Recently, SSMs [36], [37], [40], [56] have emerged as an efficient alternative, offering linear computational complexity and effective global dependence modeling compared to Transformers. To adapt Mamba to RSCD, several studies have explored architectural designs and spatiotemporal interactions. Chen et al. [44] introduced Mamba-based methods (MambaBCD, MambaSCD, and MambaBDA) that employ a visual Mamba encoder and spatiotemporal relationship modeling to capture global context and infer multitemporal changes in remote sensing imagery. To effectively model the global context of large VHR remote sensing images with linear complexity, Zhao et al. [57] proposed the RS-Mamba, which incorporates an omnidirectional selective scan module (OSSM) to address the computational challenges of dense prediction tasks. To address the limitation where existing Mamba-based methods struggle to perceive the inherent locality of change regions, Wu et al. [58] introduced CD-Lamba, which utilizes a cross-temporal locally adaptive state-space scan (CT-LASS) module to improve local–global spatiotemporal interactions. Furthermore, hybrid architectures have been proposed to compensate for the over-smoothing of local details in Mamba models. Zhang et al. [45] proposed a Mamba-based binary change detection (CDMamba) method that integrates a scaled residual ConvMamba block to jointly model global context and local details, and an adaptive global–local guided fusion module to enhance bitemporal feature interaction for accurate dense change prediction. Paranjape et al. [59] presented Mamba-based change detection (M-CD) that leverages linear-complexity sequence modeling and enhanced receptive fields to achieve more accurate segmentation of change regions. Guo et al. [60] develop attention-Mamba change detection (AM-CD), integrating an Attention-Mamba block into a multiscale dual-encoder to balance local-detail extraction with global context modeling, and further introducing a gated feature compensation difference module and a multiscale progressive decoder to reduce bitemporal feature mismatch and refine change reconstruction. Fang et al. [61] further explore Mamba-based RSCD by introducing a local–global encoder and a mask Mamba decoder, where multikernel convolutions, window attention, and mask-guided directional selective scanning are jointly employed to improve local–global fusion and boundary representation.

However, existing Mamba-based RSCD methods mainly emphasize spatial-domain sequence modeling and local–global interaction, while the role of explicit frequency-domain representation in suppressing pseudo-changes remains insufficiently investigated. To address this issue, we propose FSMamba, which incorporates a frequency-domain enhancement branch to capture bitemporal structural discrepancies and provide complementary structural cues for robust change detection.

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D. Frequency-Domain-Based RSCD Methods

Frequency-domain analysis provides a complementary perspective to spatial features, effectively characterizing global structural patterns and fine-grained intensity variations. By transforming images into the frequency domain, deep learning models can better distinguish between salient changes and background noise, leading to improved robustness in complex remote sensing scenarios. To address the neglect of frequency-domain information in existing spatial-based methods, Zhou et al. [62] proposed a joint frequency-spatial-domain network (JFSDNet), which utilizes a frequency selection module to adaptively capture frequency clues and supplement image details lost during downsampling. Tang et al. [63] proposed frequency decoupling interaction network (FDINet) to address the limitation of differentiating change types, employing a wavelet interaction module (WIM) to separate and interact with frequency components for robust fine-grained change detection. Zheng et al. [64] developed high-frequency attention network (HFA-Net) for building change detection, utilizing a high-frequency attention block (HFAB) composed of spatialwise attention and high-frequency enhancement modules to specifically highlight boundary information. Recent studies have also explored frequency-aware graph-based learning for HSI analysis. In addition, cross-modality mutual learning has been explored in HSI [65], highlighting the importance of representation alignment across heterogeneous domains. Ding et al. [66] introduce a self-supervised low-pass contrastive graph clustering (SLCGC) framework, where low-pass graph denoising is employed to suppress high-frequency noise and improve representation robustness in HSI clustering. Similarly, Ding et al. [67] propose an adaptive homophily structure graph clustering (AHSGC) that jointly models low- and high-frequency components through adaptive filtering and structure-aware graph updates. These methods demonstrate the effectiveness of frequency-aware priors in enhancing noise robustness and structural consistency.

However, these frequency-aware methods primarily address single-temporal HSI clustering and are fundamentally different from bitemporal RSCD. Despite these advances in frequency-based modeling, existing Mamba-based RSCD methods still face notable limitations, as frequency-domain information is often underexplored and rigid fusion strategies are commonly adopted, limiting their ability to effectively distinguish genuine changes from high-frequency pseudo-changes. In this article, we propose a collaborative modeling strategy to address these issues.

III. Methodology

A. Preliminaries

While SSMs provide a mathematically principled framework for sequence modeling, applying standard visual Mamba blocks directly to RSCD reveals a critical mechanistic limitation. During spatial sequence scanning, Mamba's continuous state transitions act as recursive integrators. We hypothesize that this dynamic mechanism is inherently overly sensitive to sharp, localized perturbations. Consequently, high-frequency pseudo-changes, such as misregistration errors, are easily encoded into the hidden state and erroneously propagated through the state transition matrix. Because decoupling these high-frequency noises from genuine structural variations is mathematically challenging in the spatial domain, we introduce an explicit frequency-domain prior. To avoid interfering with the delicate state evolution of SSMs, the frequency-domain filtering (FDEM) and the state-space dynamic modeling are strictly designed as parallel branches. Their representational connection is explicitly deferred to the decoding stage, which effectively compensates for Mamba's high-frequency vulnerability while preserving its pure and stable state-space dynamics.

In deep learning, SSMs [36], [37], [40], [56] provide a principled framework for modeling dynamic processes and long-range dependencies. Under the linear time-invariant (LTI) assumption, an SSM describes the transformation from an input sequence $x(t) \in \mathbb{R}$ to an output sequence $y(t) \in \mathbb{R}$ through a latent state $h(t) \in \mathbb{R}^N$, where N denotes the dimensionality of the hidden state. Rooted in classical control theory, the temporal evolution of the latent state is governed by the following linear ordinary differential equations:

$$h'(t) = \mathbf{A}h(t) + \mathbf{B}x(t) \tag{1}$$

$$y(t) = \mathbf{C}h(t) \tag{2}$$

where $\mathbf{A} \in \mathbb{R}^{N\times N}$, $\mathbf{B} \in \mathbb{R}^{N\times 1}$, and $\mathbf{C} \in \mathbb{R}^{1\times N}$ denote the state transition matrix, input projection matrix, and state-to-output projection matrix, respectively. This linear formulation characterizes the evolution of the latent state as a continuous process that accumulates historical context through the transition matrix $\mathbf{A}$ while dynamically responding to new inputs via $\mathbf{B}$. In the context of visual sequence modeling, such continuous-time representations provide a mathematically grounded mechanism for capturing temporal dependencies over long sequences.

To process discrete data, the continuous-time SSM is discretized with a fixed sampling interval Δ under the zero-order hold (ZOH) assumption. The corresponding discrete systems are computed as

$$\bar{\mathbf{A}} = e^{\Delta\mathbf{A}} \tag{3}$$

$$\bar{\mathbf{B}} = \Delta\mathbf{A}^{-1}\left(e^{\Delta\mathbf{A}} - I\right)\cdot\Delta\mathbf{B} \tag{4}$$

where $\bar{\mathbf{A}}$ and $\bar{\mathbf{B}}$ denote the discretized counterparts of $\mathbf{A}$ and $\mathbf{B}$, respectively. The discrete state update and output equations are then given by

$$h(t) = \bar{\mathbf{A}}h_{t-1} + \bar{\mathbf{B}}x_t \tag{5}$$

$$y(t) = \mathbf{C}h(t). \tag{6}$$

For an input sequence of length L_x, the output can be obtained equivalently through a global convolution operation

$$\bar{\mathbf{K}} = \left(\mathbf{C}\bar{\mathbf{B}}, \mathbf{C}\bar{\mathbf{A}}\bar{\mathbf{B}}, \ldots, \mathbf{C}\bar{\mathbf{A}}^{\mathbf{L_x-1}}\bar{\mathbf{B}}\right) \tag{7}$$

$$y = x * \bar{\mathbf{K}} \tag{8}$$

where $\bar{\mathbf{K}} \in \mathbb{R}^{\mathbf{L}_x}$ represents a structured convolution kernel derived from the state transition dynamics, and $*$ denotes the convolution operator.

Conventional spatial SSMs, such as S4 [36], adopt static parameterization, which limits their capacity for content-dependent reasoning. To overcome this limitation, the selective

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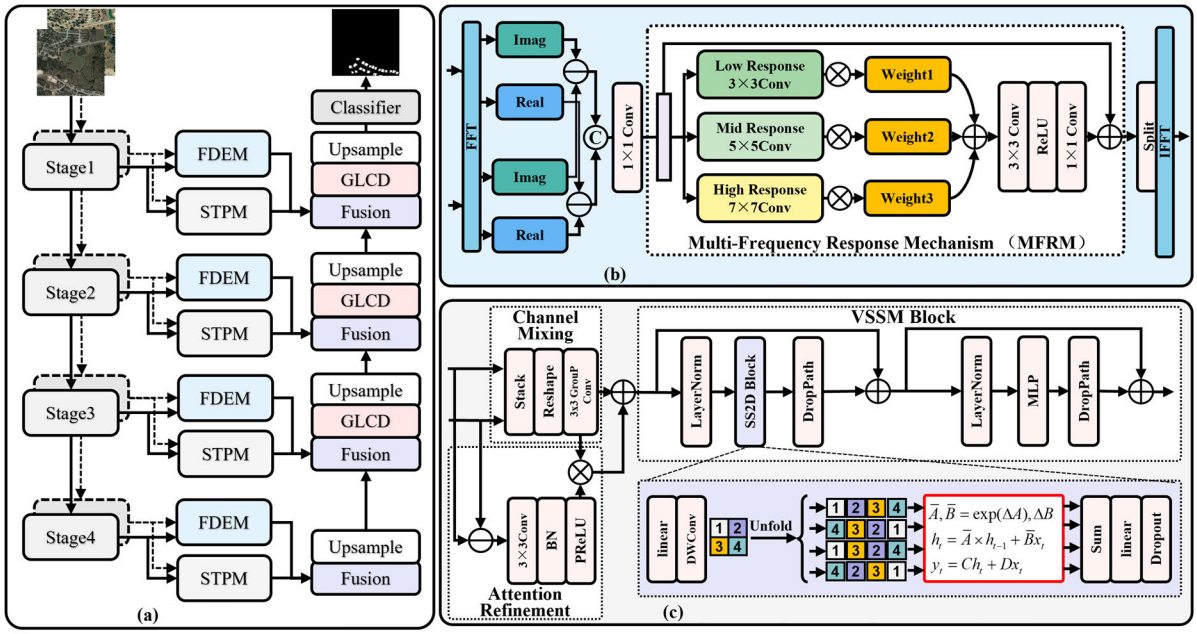


Fig. 1. Illustration of our method. (a) Overall architecture of FSMamba. (b) FDEM. (c) STPM.

SSM (S6) [37] introduces input-dependent parameters Δ, $\bar{\mathbf{A}}$, $\bar{\mathbf{B}}$, and $\mathbf{C}$, enabling dynamic state transitions conditioned on the input sequence. These parameters are computed directly from the input, allowing selective information propagation and enhanced contextual modeling. Furthermore, S6 incorporates hardware-aware optimizations for efficient GPU execution, achieving linear computational and memory complexity with respect to sequence length, which makes it particularly suitable for long-sequence visual modeling tasks.

In the proposed framework, the selective state-space mechanism is implemented within the STPM as an independent spatiotemporal modeling component. The frequency-domain enhancement branch does not directly modify the input-dependent state transition parameters of the S6. Instead, spectral filtering and selective state-space modeling operate as complementary processes, and their interaction is realized at the representation level during the subsequent feature fusion stage.

B. Overall Architecture

As illustrated in Fig. 1(a), the proposed FSMamba network introduces a unified hierarchical encoder–decoder architecture that jointly models frequency, spatial, and temporal representations to achieve fine-grained remote sensing image change detection. Specifically, FSMamba integrates three core modules to address the key limitations of existing CNN, Transformer, and Mamba-based approaches. The backbone employs SeaFormer [68] as a shared feature extractor with four hierarchical stages (Stage1–Stage4), which generate multiscale feature representations for each temporal input. At each stage, the co-registered bitemporal features are processed in parallel by the FDEM and the STPM. FDEM operates in the frequency domain to suppress nonstructural high-frequency pseudo-changes and enhance structure-consistent discrepancy cues, providing robust structural representations. In contrast, STPM performs Mamba-based selective state-space modeling in the spatial–temporal domain to capture long-range dependencies and temporal consistency through input-dependent state transitions. It is important to note that FDEM and STPM are designed as complementary modeling branches rather than explicitly coupled mechanisms. FDEM focuses on spectral-structural robustness, while STPM emphasizes dynamic contextual modeling via selective state evolution. Their interaction is realized during the decoding stage, where the GLCD adaptively fuses frequency-enhanced structural features and selectively modeled spatiotemporal representations through an attention-guided global–local interaction mechanism. This collaborative integration enables simultaneous preservation of large-scale semantic consistency and fine-grained boundary details. The final change detection results are derived from a classifier following the decoder output.

C. Frequency-Domain Enhancement Module

Accurate change detection in VHR remote sensing imagery is often hindered by pseudo-changes and subtle structural inconsistencies caused by illumination variations and geometric misregistration. To mitigate these challenges, we propose an FDEM [Fig. 1(b)] that explicitly models spectral–structural discrepancies between bitemporal feature maps through Fourier-based representation learning. By transforming the feature maps into the frequency domain and enhancing the key frequency components, FDEM helps suppress irrelevant variations and highlight structural differences.

Given two feature maps $X_A \in \mathbb{R}^{C\times H\times W}$ and $X_B \in \mathbb{R}^{C\times H\times W}$ extracted from the SeaFormer [68] backbone, FDEM transforms them into the frequency domain using the 2-D fast Fourier transform (FFT)

$$F_A = FFT(X_A) \tag{9}$$

$$F_B = FFT(X_B). \tag{10}$$

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The Fourier transform decomposes the input feature maps into complex-valued representations, capturing both amplitude and phase information across the frequency spectrum. This transformation allows the model to focus on global structures and patterns within the images, rendering it less sensitive to local perturbations such as illumination variations and misregistration errors. To model the structural differences between the two feature maps, FDEM computes the complex difference between the two frequency-domain representations

$$\Delta F = Conv_{1\times1}\left(Concat\left(F_A - F_B\right)\right). \tag{11}$$

This step eliminates nonstructural pseudo-changes and emphasizes frequency-domain cues for true semantic changes. To enhance the discriminative capability of frequency-domain features, FDEM incorporates a multifrequency response mechanism (MFRM) that learns frequency-dependent attention weights

$$w_i = W \cdot \Delta F \tag{12}$$

$$W(\cdot) = Sigmoid\left(Conv_{7\times7}(\cdot) + Conv_{5\times5}(\cdot) + Conv_{3\times3}(\cdot)\right). \tag{13}$$

The multiscale convolutions capture local (3×3), middle (5×5), and global (7×7) spectral dependencies, producing a frequency-response map w_i that quantifies the relative importance of different frequency components. The enhanced spectral representation, which serves as the input for the multiscale spectral fusion stage, is then obtained as

$$F_{enhanced} = \sum_{i=1}^{3} w_i \cdot \left(M_i \odot \Delta F\right) \tag{14}$$

where $\odot$ denotes the elementwise multiplication and M_i is represented as binary masks. After frequency-domain enhancement, the processed spectral features are connected and transformed back into the spatial domain via the convolution and inverse FFT (IFFT)

$$F_{out} = \left(Conv_{1\times1}\left(ReLU\left(Conv_{3\times3}\left(F_{enhanced}\right)\right)\right)\right) + \Delta F \tag{15}$$

$$F_{FDEM} = IFFT\left(Split\left(F_{out}\right)\right). \tag{16}$$

By reshaping the spectral statistics of bitemporal features, FDEM provides robust, frequency-consistent representations that serve as complementary structural priors for the subsequent global–local decoding stage.

D. Spatiotemporal Perception Module

To effectively capture dynamic variations between bitemporal feature representations, we propose an STPM [Fig. 1(c)], which jointly models cross-temporal dependencies and spatially localized changes. Unlike conventional convolution-based fusion strategies that rely on concatenation or differencing, STPM performs progressive spatiotemporal interaction through three functional stages: channel mixing, attention refinement, and VSSM block. Operating in parallel with the FDEM, STPM directly processes the spatial-domain bitemporal features to capture input-dependent selective state-space dynamics. This hierarchical design enables the network to perceive both localized structural differences and long-range temporal coherence independently of the frequency branch.

Specifically, given two temporal feature maps $X_A, X_B \in \mathbb{R}^{C\times H\times W}$, STPM first employs a channel mixing block to achieve spectral fusion across temporal inputs. Concretely, the two feature maps are first stacked and reshaped along the channel dimension to form a unified $2C$-channel representation. A 3×3 group convolution is then applied, enabling efficient cross-temporal channel interaction while preserving spatial alignment

$$F_{mix} = \sigma\left(BN\left(GConv_{3\times3}^{(g)}\left(Reshape\left(Stack\left(X_A, X_B\right)\right)\right)\right)\right) \tag{17}$$

where $Stack(\cdot)$ denotes stacking along a new temporal dimension, $Reshape(\cdot)$ reorganizes the stacked tensor into a $2C$-channel feature map, $GConv_{3\times3}^{(g)}(\cdot)$ denotes a 3×3 group convolution, $BN(\cdot)$ represents the batch normalization, and $\sigma(\cdot)$ denotes the rectified linear unit (ReLU) activation function. This operation captures spectral correlations and shared contextual cues between the bitemporal features, forming the primary pathway for cross-temporal interaction.

To further enhance discriminative change perception, an attention refinement block is introduced. The absolute difference between the two temporal features is first computed and then projected through a 3×3 convolution, followed by batch normalization and PReLU activation:

$$F_d = PReLU\left(BN\left(Conv_{3\times3}\left(|X_A - X_B|\right)\right)\right) \tag{18}$$

where $Conv_{3\times3}$ denotes a 3×3 convolution layer, $BN(\cdot)$ represents the batch normalization, and PReLU denotes the parametric ReLU activation function. The refined difference map is subsequently fused with the mixed features via elementwise modulation

$$F_{fuse} = F_{mix} + F_{mix} \odot F_d \tag{19}$$

where $\odot$ denotes the elementwise multiplication. This adaptive interaction selectively amplifies informative temporal variations while suppressing noise-dominated responses and pseudo-changes induced by environmental disturbances.

After local refinement, the fused representation X_f is passed through a VSSM block to model long-range dependencies and temporal consistency in a state-space manner

$$F_{SS2D} = DropPath\left(SS2D\left(LN\left(F_{fuse}\right)\right)\right) + F_{fuse}. \tag{20}$$

By leveraging the selective state-space mechanism, the VSSM block serves as a global context integrator, enabling implicit sequential reasoning across spatial positions and temporal dimensions. In particular, the core 2-D-selective-scan (SS2D) block adopts a four-directional selective scanning strategy (left-to-right, right-to-left, top-to-bottom, and bottom-to-top) to transform 2-D feature maps into multiple 1-D sequences for state-space propagation. This design aggregates global spatial context from complementary directions with linear complexity, which is crucial for preserving structural continuity while suppressing transient variations.

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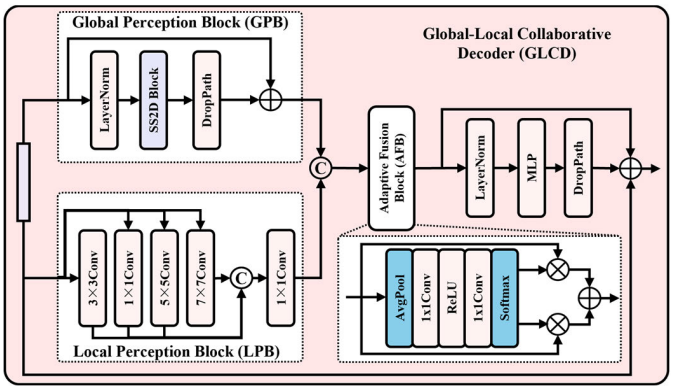


Fig. 2. Global-local collaborative decoder (GLCD).

Finally, a residual convolutional head is employed to aggregate the enhanced spatiotemporal information and stabilize network optimization

$$F_{STPM} = DropPath\left(MLP\left(LN\left(F_{SS2D}\right)\right)\right) + F_{SS2D}. \quad (21)$$

Through the synergistic combination of spectral channel mixing, attention-guided difference enhancement, and global state-space modeling, STPM achieves an effective balance between local sensitivity and temporal stability, enabling robust discrimination of genuine structural changes from transient ones.

By jointly aggregating frequency-enhanced structural representations from FDEM and selectively modeled spatiotemporal features from STPM, the proposed decoder enables complementary integration between spectral filtering and input-dependent state-space modeling at the representation level.

E. Global–Local Collaborative Decoder

To effectively integrate the heterogeneous representations learned by the encoder and progressively restore spatial structures in postchange regions, we propose the GLCD (Fig. 2). While the encoder extracts discriminative cues in both frequency and spectral–temporal domains via the FDEM and STPM, the decoder is responsible for reconstructing change regions with structural continuity and boundary precision. Conventional decoders often treat multidomain features independently, which may lead to spatial inconsistencies. To address this issue, GLCD adopts a hierarchical attention-guided VSSM strategy to perform adaptive global–local feature fusion.

Formally, the GLCD receives the frequency-enhanced feature F_{FDEM} from the FDEM and the spatiotemporal interaction feature F_{STPM} from the STPM. These heterogeneous features are first concatenated along the channel dimension and projected into a unified latent embedding $\mathbf{Z}$ via a 1×1 convolution. The latent representation is then processed by two complementary branches

$$F_{GPB} = DropPath\left(SS2D\left(LN\left(\mathbf{Z}\right)\right)\right) \quad (22)$$

$$F_{LPB} = Conv_{1\times1}\left(Concat\left[Conv_{3\times3,1\times1,5\times5,7\times7}\left(\mathbf{Z}\right)\right]\right) \quad (23)$$

where F_{GPB} captures long-range scene context using the SS2D mechanism, and F_{LPB} preserves fine-grained structural details via convolutions.

To dynamically balance these conflicting yet complementary cues, we design a branchwise attention guidance mechanism named adaptive fusion block (AFB). The global and local features are first concatenated along the channel dimension. A global context descriptor is then obtained via global average pooling (GAP) and mapped to adaptive branch weights using a lightweight perception network

$$\omega_{global}, \omega_{local} = Soft\max\left(\psi_{mlp}\left(GAP\left(\left[F_{GPB}, F_{LPB}\right]\right)\right)\right) \quad (24)$$

where $[\cdot]$ denotes the concatenation, and ψ_{mlp} consists of two 1×1 convolutional layers with an ReLU activation. The Softmax function ensures that $\omega_{global} + \omega_{local} = 1$, enforcing a competitive relationship between the global and local views. Finally, the output feature F_{GLCD} is generated by the weighted aggregation of both branches

$$F_{GLCD} = \omega_{global} \cdot F_{GPB} + \omega_{local} \cdot F_{LPB}. \quad (25)$$

This mechanism enables the decoder to adaptively emphasize global semantic consistency and local boundary accuracy based on the image content, thereby facilitating change reconstruction.

F. Loss Function

In RSCD scenarios, the quantity of unchanged pixels significantly preponderates over that of changed pixels. This severe class imbalance frequently induces the network to overfit to the background, consequently resulting in the omission of small targets and the blurring of object boundaries. To mitigate this issue, we propose a hybrid loss function that synergizes the advantages of Focal Loss [69] and Dice Loss [70] to supervise the network training. Specifically, the Focal Loss is utilized to mine hard examples and alleviate the dominance of background samples, while the Dice Loss is incorporated to optimize the intersection-over-union (IoU) metric, thereby ensuring region-level structural consistency. The total optimization objective, denoted as L_{total}, is formulated as a weighted combination of these two components

$$L_{total} = \lambda_1 L_{focal} + \lambda_2 L_{dice} \quad (26)$$

where λ_1 and λ_2 represent the coefficients of the loss function.

IV. Experiments

To comprehensively evaluate the effectiveness of the proposed FSMamba framework, extensive experiments are conducted on five representative benchmark datasets for RSCD. This section first introduces the employed datasets, followed by the implementation details, quantitative and qualitative comparisons with state-of-the-art methods, and ablation studies to verify the contribution of each component in the proposed framework.

A. Dataset Introduction

We evaluate the proposed method on five widely used public change detection datasets, including LEVIR-CD [48], WHU-CD [71], SYSU-CD [50], DSIFN-CD [20], and cropland change detection (CLCD) [53], which cover diverse spatial resolutions, scene complexities, and change types.

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1) LEVIR-CD Dataset: The LEVIR-CD dataset [48] consists of 637 pairs of very high-resolution Google Earth images with a spatial resolution of 0.5 m/pixel and temporal intervals ranging from 5 to 14 years. The dataset mainly focuses on building-related changes, such as building construction and demolition, and contains more than 31 000 annotated change instances. The original images of size 1024×1024 are cropped into nonoverlapping patches of 256×256, resulting in 7120, 1024, and 2048 samples for training, validation, and testing, respectively.

2) WHU-CD Dataset: The WHU-CD dataset [71] is designed for building change detection and is collected over Christchurch, New Zealand, using aerial imagery acquired in April 2012 and April 2016. The dataset covers an area of approximately 20.5 km^2, with very high spatial resolutions ranging from 0.2 to 0.3 m/pixel. It focuses on building-related changes, where the number of buildings increases from 12 796 in 2012–16 077 in 2016. The images are cropped into nonoverlapping patches of 256×256 and divided into 6096, 762, and 762 samples for training, validation, and testing, respectively.

3) SYSU-CD Dataset: The SYSU-CD dataset [50] contains 20 000 pairs of bitemporal aerial images acquired over Hong Kong between 2007 and 2014, with a spatial resolution of 0.5 m/pixel and a fixed patch size of 256×256. The dataset includes various change types, such as building construction, suburban expansion, groundwork before construction, vegetation variation, road expansion, and sea construction, involving multiple object categories, including buildings, roads, vegetation, and ships. The dataset is divided into 12 000, 4000, and 4000 image pairs for training, validation, and testing, respectively.

4) CLCD Dataset: The CLCD dataset [53] consists of 600 pairs of bitemporal images acquired by the Gaofen-2 satellite over Guangdong Province, China, in 2017 and 2019. The images have a spatial size of 512×512, with resolutions ranging from 0.5 to 2 m/pixel. The dataset mainly annotates cropland-related changes, including buildings, roads, lakes, and bare land. A total of 320, 120, and 120 image pairs are used for training, validation, and testing, respectively.

5) DSIFN-CD Dataset: The DSIFN-CD [20] is a public binary change detection dataset composed of bitemporal images collected over six major cities in China, with a spatial resolution of 2 m/pixel. The dataset involves diverse land-cover changes, including buildings, roads, and farmland. The images are cropped into patches of size 512×512, yielding 3600, 340, and 48 image pairs for training, validation, and testing, respectively.

B. Evaluation Metrics

To quantitatively evaluate the performance of the proposed FSMamba framework and the compared methods, five widely adopted evaluation metrics are employed, including precision (Pre.), recall (Rec.), IoU, overall accuracy (OA), and $F1$-score ($F1$). These metrics are used to comprehensively assess the classification reliability of both changed and unchanged regions on all test datasets.

Specifically, precision measures the proportion of correctly predicted changed pixels among all predicted changed pixels, while recall reflects the ability of a model to correctly identify actual changed pixels. IoU evaluates the overlap ratio between the predicted change regions and the ground-truth change regions. OA represents the proportion of correctly classified pixels over the entire image, and the $F1$-score, as the harmonic mean of precision and recall, provides a balanced evaluation of change detection performance, especially under class-imbalanced conditions.

The above metrics are mathematically defined as follows:

$$\mathrm{Pr}\,ecision = \frac{TP}{TP + FP} \tag{27}$$

$$Recall = \frac{TP}{TP + FN} \tag{28}$$

$$IoU = \frac{TP}{TP + FN + FP} \tag{29}$$

$$OA = \frac{TP + TN}{TP + TN + FP + FN} \tag{30}$$

$$F1 = \frac{2 \times (Precision \times Recall)}{\mathrm{Pr}\,ecision + Recall} \tag{31}$$

where TP, TN, FP, and FN denote the numbers of true positive, true negative, false positive, and false negative pixels, respectively.

C. Experimental Step

All experiments were implemented using the PyTorch framework and conducted on an NVIDIA RTX 3090 GPU. The proposed model was trained for 100 epochs with an initial learning rate of 0.01 and optimized using stochastic gradient descent (SGD) with a momentum of 0.9 and a weight decay of 5×10^{-4}. To achieve a balance between convergence speed and training stability, a stepwise learning rate decay strategy (StepLR) was employed, in which the learning rate was reduced by a fixed factor every 50 epochs. The batch size was set to 4 for all experiments. To improve model generalization, data augmentation techniques including random horizontal and vertical flipping, as well as random Gaussian blurring, were applied during training. Specifically, the Gaussian blur was implemented via the scikit-image library, with the standard deviation (σ) uniformly sampled from [0.15, 1.30] and a dynamically adapted filter window. This configuration explicitly forces the network to suppress frequency-domain perturbations without the need for synthetic noise injection. Furthermore, with the SeaFormer [68] backbone initialized using pretrained weights, all modules of the proposed architecture were jointly optimized in an end-to-end manner. For the decoder architecture, the channel dimensions were set to [64, 128, 192, 256], with a uniform decoding depth of [4, 4, 4, 4] across all hierarchical levels, ensuring consistent feature extraction and multiscale feature fusion. Finally, the overall loss function was composed of Focal Loss and Dice Loss with equal weights ($\lambda_1 = \lambda_2 = 0.5$). Beyond the experimental validation provided by our ablation studies (see Table S3), this balanced weighting theoretically ensures stable training convergence: gradient fluctuations potentially induced by Focal Loss during hard-example mining are effectively

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TABLE I

COMPARISON RESULTS ON LEVIR-CD, WHU-CD, AND SYSU-CD DATASETS. THE TOP THREE RESULTS ARE HIGHLIGHTED IN RED, GREEN, AND BLUE, RESPECTIVELY

Types	Models	LEVIR-CD					WHU-CD					SYSU-CD				
		F1	Pre.	Rec.	IoU	OA	F1	Pre.	Rec.	IoU	OA	F1	Pre.	Rec.	IoU	OA
CNN -based	FC-EF [18]	83.4	86.91	80.17	71.53	98.39	72.01	77.69	67.10	56.26	92.07	72.57	72.28	72.85	56.94	87.01
	FC-Siam-Diff [18]	86.31	89.53	83.31	75.92	98.67	58.81	47.33	77.66	41.66	95.63	64.08	84.94	51.45	47.15	86.40
	FC-Siam-Conc [18]	83.69	91.99	76.77	71.96	98.49	66.63	60.88	73.58	49.95	97.04	76.89	83.03	71.62	62.45	89.35
	IFNet [20]	88.13	**94.02**	82.93	78.77	98.87	83.40	**96.91**	73.19	71.52	98.83	76.53	79.59	73.58	61.91	89.17
	DTCDSCN [72]	87.67	88.53	86.83	78.05	98.77	71.95	63.92	82.30	56.19	97.42	80.11	83.19	77.25	66.82	90.96
	SNUNet [19]	88.16	89.18	87.17	78.83	98.82	83.50	85.60	81.49	71.67	98.71	77.43	81.93	73.39	63.17	89.91
	LGPNet [73]	89.37	**93.07**	85.95	80.78	99.00	79.75	89.68	71.81	66.33	98.33	77.63	80.97	74.56	63.44	89.87
	USSFC-Net [74]	88.80	87.18	90.49	79.86	98.84	88.93	91.56	86.43	80.06	99.01	81.05	81.05	81.05	68.13	91.06
	AFCF3D-Net [75]	90.44	91.18	89.72	82.55	99.03	92.07	93.56	90.62	85.30	99.28	82.15	84.22	80.18	69.71	91.78
	CDXLSTM [76]	**90.89**	91.52	90.27	**83.30**	**99.07**	92.58	93.71	91.49	86.19	99.33	**87.96**	**88.17**	**87.76**	**79.10**	91.37
Transformer -based	BIT [29]	89.31	89.24	89.37	80.68	98.92	83.98	86.64	81.48	72.39	98.75	79.24	84.89	74.29	65.62	90.82
	ChangeFormer [30]	90.40	92.05	88.80	82.48	99.04	81.82	87.25	77.03	69.24	94.80	78.21	79.37	77.08	64.22	89.87
	SARASNet [77]	90.44	91.42	89.48	82.55	**99.11**	89.55	88.68	90.44	81.08	99.05	79.24	86.35	73.21	65.62	91.08
	DMINet [52]	90.46	92.16	88.83	82.59	99.04	93.43	94.89	**92.02**	87.68	**99.48**	82.11	84.99	79.41	69.95	**91.84**
	BiFA [31]	90.69	91.52	89.86	82.96	**99.06**	**94.37**	95.15	**93.60**	**89.34**	**99.56**	84.12	86.98	81.44	72.59	**92.75**
Mamba -based	ChangeMamba [44]	90.51	90.73	90.30	82.67	99.04	90.08	92.94	87.38	81.95	99.12	79.38	82.43	76.54	65.80	90.62
	CDMamba [45]	90.75	91.43	90.08	83.07	**99.06**	**93.76**	**95.58**	92.01	**88.26**	**99.51**	**86.85**	**87.85**	**85.96**	**77.47**	90.75
	RS-Mamba [57]	90.67	90.70	**90.63**	82.93	99.05	91.50	93.21	89.85	84.33	99.23	75.70	78.20	73.35	60.90	88.89
	CD-Lamba [58]	**93.45**	93.03	**95.07**	**89.74**	98.87	92.51	93.45	91.59	86.07	99.32	82.66	81.66	83.68	70.44	91.72
	Ours	**95.37**	**95.55**	**95.19**	**95.37**	**99.11**	**95.68**	**95.69**	**95.67**	**92.00**	99.34	**88.98**	**89.56**	**88.43**	**80.66**	**92.17**

TABLE II

COMPARISON RESULTS ON CLCD AND DSIFN-CD DATASETS. THE TOP THREE RESULTS ARE HIGHLIGHTED IN RED, GREEN, AND BLUE, RESPECTIVELY

Types	Models	CLCD					DSIFN-CD				
		F1	Pre.	Rec.	IoU	OA	F1	Pre.	Rec.	IoU	OA
CNN -based	FC-EF [18]	48.64	73.34	36.29	32.14	94.30	59.71	61.80	57.75	42.56	86.77
	FC-Siam-Diff [18]	44.10	72.97	31.60	28.29	94.04	62.95	68.44	58.27	45.93	88.35
	FC-Siam-Conc [18]	54.48	68.21	45.22	37.35	94.35	60.88	59.08	62.80	43.76	86.30
	IFNet [20]	48.65	49.96	47.41	32.14	92.55	60.10	67.86	53.94	42.96	87.83
	DTCDSCN [72]	60.13	62.98	57.53	42.99	94.32	63.72	53.87	77.99	46.76	84.91
	SNUNet [19]	66.70	73.76	60.88	50.04	95.48	66.18	60.60	72.89	49.45	87.34
	LGPNet [73]	63.03	70.54	56.96	46.01	95.03	63.20	49.96	**85.97**	46.19	82.99
	USSFC-Net [74]	63.04	64.83	61.34	46.03	94.42	65.00	62.03	68.28	48.15	87.65
	AFCF3D-Net [75]	76.92	**84.20**	70.79	62.49	**96.84**	71.44	67.21	76.25	55.57	**89.64**
	CDXLSTM [76]	**78.73**	83.15	74.76	64.92	**96.99**	**81.88**	**81.29**	**82.51**	**71.03**	89.56
Transformer -based	BIT [29]	67.10	73.07	62.04	50.49	95.47	69.26	68.36	70.18	52.97	89.41
	ChangeFormer [30]	58.44	65.00	53.07	41.28	94.38	64.52	57.90	72.85	47.63	87.72
	SARASNet [77]	74.70	76.68	72.83	59.62	96.33	67.49	68.18	66.81	50.93	88.96
	DMINet [52]	71.64	73.24	70.11	55.81	95.87	68.13	60.17	78.51	51.66	87.79
	BiFA [31]	**84.02**	**84.02**	**74.98**	**65.61**	**97.08**	71.34	**73.99**	68.87	55.45	**90.80**
Mamba -based	ChangeMamba [44]	70.00	75.39	65.33	53.85	95.83	65.91	70.60	61.80	49.15	89.14
	CDMamba [45]	71.58	79.48	69.05	56.02	95.96	-	-	-	-	-
	RS-Mamba [57]	71.27	72.95	69.67	**72.95**	95.82	64.79	53.58	**81.93**	47.92	84.87
	CD-Lamba [58]	78.06	79.20	**76.96**	64.02	96.78	**71.66**	67.88	75.89	**55.84**	**89.80**
	Ours	**87.74**	**89.11**	**86.48**	**79.73**	96.74	**80.35**	**77.83**	**84.27**	**68.82**	87.57

regularized by the bounded region-level optimization of Dice Loss, preventing gradient domination and yielding a smooth and stable loss descent.

D. Performance Comparison

To comprehensively evaluate the proposed model, it is systematically compared with a broad range of state-of-the-art change detection methods, which are categorized into three architectural paradigms: CNN-based, Transformer-based, and Mamba-based approaches. The CNN-based methods include FC-EF [18], FC-Siam-Diff [18], FC-Siam-Conc [18], IFNet [20], DTCDSCN [72], SNUNet [19], LGPNet [73], USSFC-Net [74], AFCF3D-Net [75], and CDXLSTM [76]; the Transformer-based methods comprise BIT [29], ChangeFormer [30], SARASNet [77], DMINet [52], and BiFA [31]; and the Mamba-based methods include ChangeMamba [44], CDMamba [45], RS-Mamba [57], and CD-Lamba [58]. All methods are evaluated under identical experimental settings to ensure a fair comparison. Performance is assessed from three complementary perspectives: quantitative results, qualitative visual analysis, and model efficiency.

1) Quantitative Results: The quantitative comparison results are summarized in Tables I and II, where the best, second-best, and third-best performances are highlighted in red, green, and blue, respectively. Overall, our approach consistently achieves top-ranked performance across most metrics and datasets, demonstrating clear advantages over existing state-of-the-art change detection methods.

On the LEVIR-CD dataset, our approach outperforms all competing methods across the major evaluation metrics. In comparison with the Mamba-based baseline, CD-Lamba, it improves the IoU by 5.63 percentage point (pp), together with gains of 2.52 pp in precision and 0.12 pp in recall, indicating enhanced spatial consistency and boundary accuracy of the detected change regions while preserving balanced detection sensitivity. In addition, an OA of 99.11% is achieved, exceeding that of CD-Lamba by 0.24 pp, which confirms the robustness of the proposed design. This advantage can be directly attributed to the dataset's focus on building-level changes, which exhibit regular but fine-grained structural patterns. Specifically, the proposed GLCD module adaptively fuses multiscale features, allowing the network to preserve these sharp, complex building boundaries, while the FDEM effectively filters out localized high-frequency noise in homogeneous backgrounds.

Consistent performance gains are also observed on the WHU-CD dataset. This method achieves the best overall results, with an $F1$-score of 95.68%, an IoU of 92.00%, and an OA of 99.34%, while maintaining a well-balanced precision

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(95.69%) and recall (95.67%). Compared with the second-best model, BiFA, improvements of 1.31 pp in $F1$-score, 2.07 pp in recall, and 2.66 pp in IoU are obtained, demonstrating a stronger capability in detecting change regions and achieving more accurate spatial overlap. Given the dense urban scenes and frequent illumination/shadow variations inherent in the WHU-CD dataset, the superior precision and $F1$-score stem fundamentally from the FDEM. By operating in the frequency domain, FDEM successfully suppresses the high-frequency pseudo-changes caused by architectural shadows and viewing geometry discrepancies, enabling the model to focus on genuine structural semantics.

On the more challenging SYSU-CD dataset, the proposed approach continues to exhibit superior performance, reporting an $F1$-score of 88.98% and an IoU of 80.66%, with balanced precision (89.56%) and recall (88.43%). Relative to the CNN-based baseline CDXLSTM, the $F1$-score and IoU are improved by 1.02 and 1.56 pp, respectively. Moreover, compared with the strongest Mamba-based competitor, CDMamba, further gains of 2.13 pp in $F1$-score and 3.19 pp in IoU are achieved. The SYSU-CD dataset is uniquely characterized by highly diverse change types (e.g., suburban expansion, vegetation variation, and sea construction) and extremely complex backgrounds. The consistent benefit of FSMamba in $F1$-score and IoU demonstrates that the synergy between our parallel modules can effectively handle high intraclass variance. Specifically, the STPM dynamically captures distinct spatiotemporal contexts for varying land covers, while the FDEM robustly decouples genuine structural variations from heterogeneous background clutter, minimizing false alarms in challenging urban-to-suburban transitions.

For large-scale and heterogeneous scenarios represented by the CLCD dataset, our approach again delivers the best overall performance, achieving an $F1$-score of 87.74% and an IoU of 79.73%, supported by high precision (89.11%) and recall (86.48%). The CLCD dataset predominantly features large-scale, irregularly shaped cropland changes with massive seasonal spectral shifts. The substantial performance gap between FSMamba and competing methods in this scenario validates the distinct advantages of our architecture. On the one hand, the STPM's linear-complexity selective scanning is exceptionally well-suited for capturing the long-range continuous contexts required for massive cropland regions. On the other hand, the FDEM inherently isolates and suppresses nonsemantic spectral variations caused by seasonal crop changes, translating into the remarkable 15.71-pp IoU improvement over the Mamba-based baseline CD-Lamba.

On the DSIFN-CD dataset, this method achieves competitive results, with an $F1$-score of 80.35% and the highest IoU of 68.82%, together with precision and recall of 77.83% and 84.27%, respectively. Although slightly lower $F1$-score (-1.53 pp) and IoU (-2.21 pp) are observed compared with the best CNN-based baseline CDXLSTM, the proposed approach substantially outperforms the strongest Mamba-based method CD-Lamba, achieving improvements of 8.69 pp in $F1$-score and 12.98 pp in IoU. This performance gap may be attributed to the characteristics of the DSIFN-CD dataset, which primarily involves large-scale and explicit semantic changes with relatively limited fine-grained structural variations, thereby favoring long short-term memory (LSTM)-based models that excel at region-level temporal modeling. Overall, the results demonstrate that the proposed method remains competitive and effective. The competitive performance on DSIFN-CD further confirms that even in diverse, large-scale urban transformations, the proposed frequency-enhanced representations provide reliable discriminative cues, effectively separating genuine land-cover evolution from the complex surrounding urban noise.

2) Qualitative Results: To further validate the effectiveness of the proposed approach, qualitative comparisons are conducted on representative test samples from LEVIR-CD, WHU-CD, SYSU-CD, CLCD, and DSIFN-CD, as illustrated in Figs. 3–7. A consistent color-coding scheme is adopted for visualization, where TPs are shown in white, TNs in black, FPs in red, and FNs in green, enabling an intuitive and comprehensive inspection of change detection performance. As shown in the figures, the proposed approach consistently generates more complete, spatially coherent, and visually cleaner change maps across all datasets. On LEVIR-CD, which mainly contains building-level changes with clear structural patterns, the proposed approach effectively preserves fine-grained object boundaries while significantly reducing false alarms in homogeneous background regions. This results in more accurate delineation of building changes, especially along complex edges.

For WHU-CD and SYSU-CD, both of which involve complex urban environments with diverse land-cover types and dense scene structures, the proposed approach demonstrates a strong capability to suppress spurious detections and recover continuous change regions. Compared with competing methods, fewer fragmented regions and misclassified pixels are observed, indicating improved robustness against background clutter and class imbalance. In particular, FPs around roads, shadows, and densely built-up areas are effectively reduced, while FNs within large changed regions are minimized.

On the CLCD dataset, characterized by large-scale scenes and heterogeneous land-cover changes, the proposed approach exhibits superior robustness. As illustrated in Fig. 6, it produces cleaner change maps with more consistent spatial structures and sharper boundaries, while substantially reducing noise and scattered false detections that commonly appear in other methods.

For the DSIFN-CD dataset, which includes diverse change patterns and challenging background variations, the proposed approach maintains competitive visual quality. Although some competing methods yield comparable results in certain regions, clearer separation between changed and unchanged areas is achieved, with improved boundary localization and fewer isolated false detections.

3) Model Efficiency: To comprehensively evaluate computational efficiency, we benchmarked FSMamba against representative CNN, Transformer, and Mamba-based methods at 256×256 and 512×512 resolutions. As reported in Table III, FSMamba exhibits very low computational complexity, requiring only 45.74 giga floating-point operations (GFLOPs) at 512×512 resolution, which is approximately 8% of ChangeFormer's computational cost and substantially lower than the CNN

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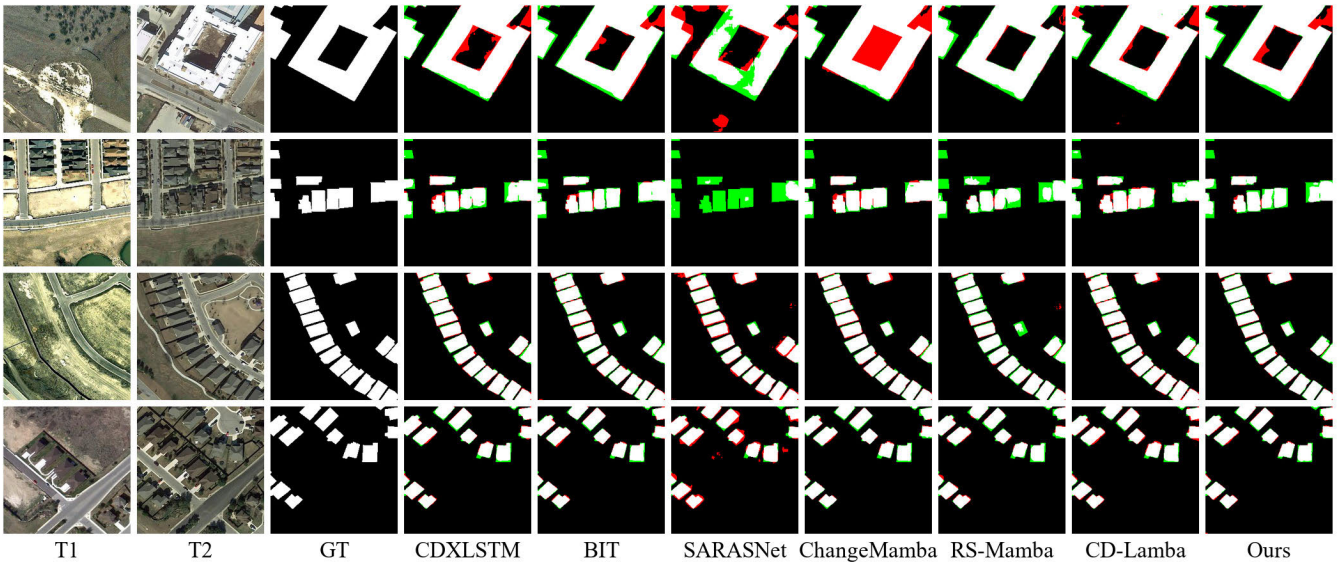


Fig. 3. Visualization results of different methods on the LEVIR-CD test set. TPs are shown in white, TNs in black, FPs in red, and FNs in green.

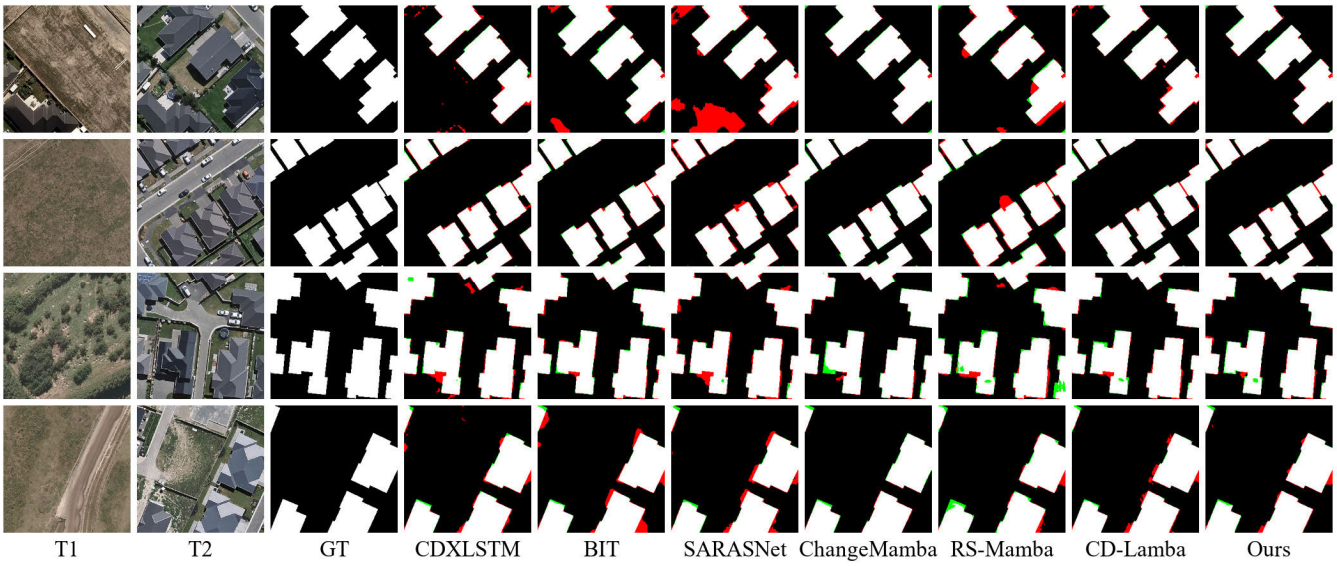


Fig. 4. Visualization results of different methods on the WHU-CD test set. TPs are shown in white, TNs in black, FPs in red, and FNs in green.

baseline SNUNet. Benefiting from the linear-complexity selective state-space modeling mechanism, FSMamba demonstrates favorable scalability when increasing input resolution. When the spatial resolution is doubled (from 256 × 256–512 × 512), the GFLOPs increase from 14.03 to 45.74, while the inference latency increases from 51.79 to 57.42 ms. Despite the substantial growth in theoretical computation, the runtime increase remains moderate in practice due to efficient GPU parallelization. Compared with the Mamba-based baseline CD-Lamba, FSMamba achieves significantly faster inference (57.42 versus 119.87 ms at 512 × 512) and lower GPU memory consumption (549.64 versus 691.34 MB), while simultaneously delivering higher detection accuracy. These results indicate that FSMamba achieves a favorable balance between computational efficiency and practical deployability for large-scale RSCD.

E. Ablation Studies

To systematically evaluate the contribution of individual components and design choices in FSMamba, a series of ablation experiments is conducted on the CLCD dataset. The corresponding analyses are presented as follows.

1) Effects of Different Blocks: To evaluate the contribution of the core modules in FSMamba, ablation experiments are conducted on the CLCD dataset by individually removing the main blocks, including FDEM, STPM, and GLCD. As reported in Table IV, all variants without a main block exhibit performance degradation compared with the complete model, confirming that each component contributes to the overall effectiveness. Among them, removing FDEM leads to the largest decrease, with the $F1$-score dropping from 87.74% to 86.90% and OA decreasing from 96.74% to 95.81%, suggesting that the frequency-domain branch provides beneficial structural cues for discriminative land-cover change

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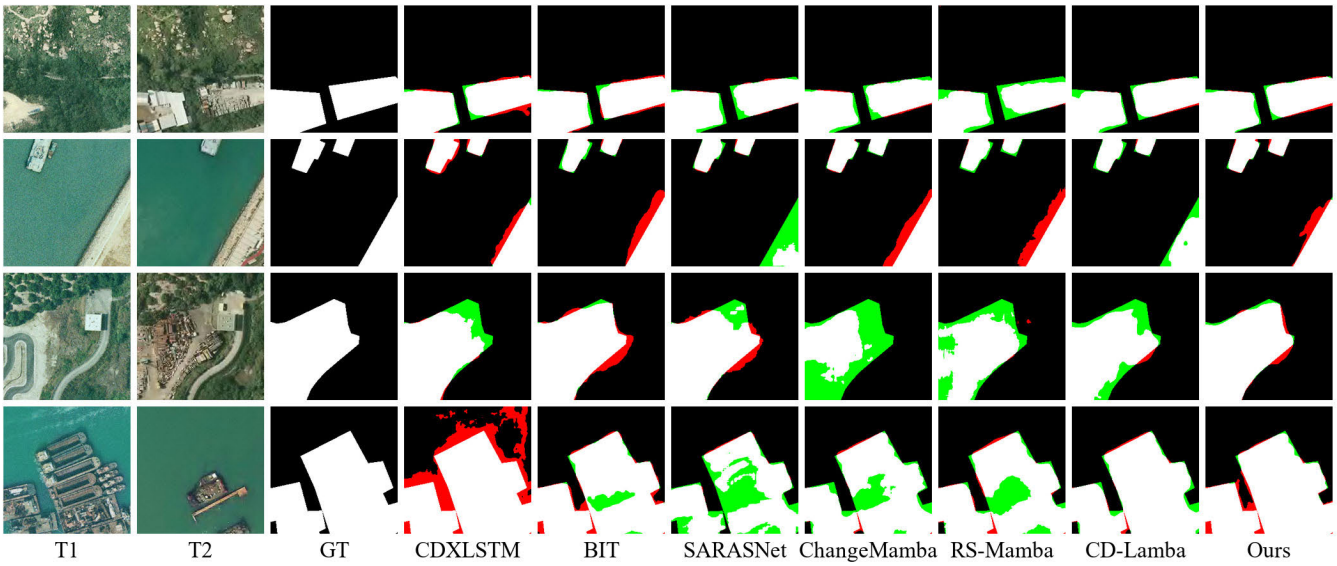


Fig. 5. Visualization results of different methods on the SYSU-CD test set. TPs are shown in white, TNs in black, FPs in red, and FNs in green.

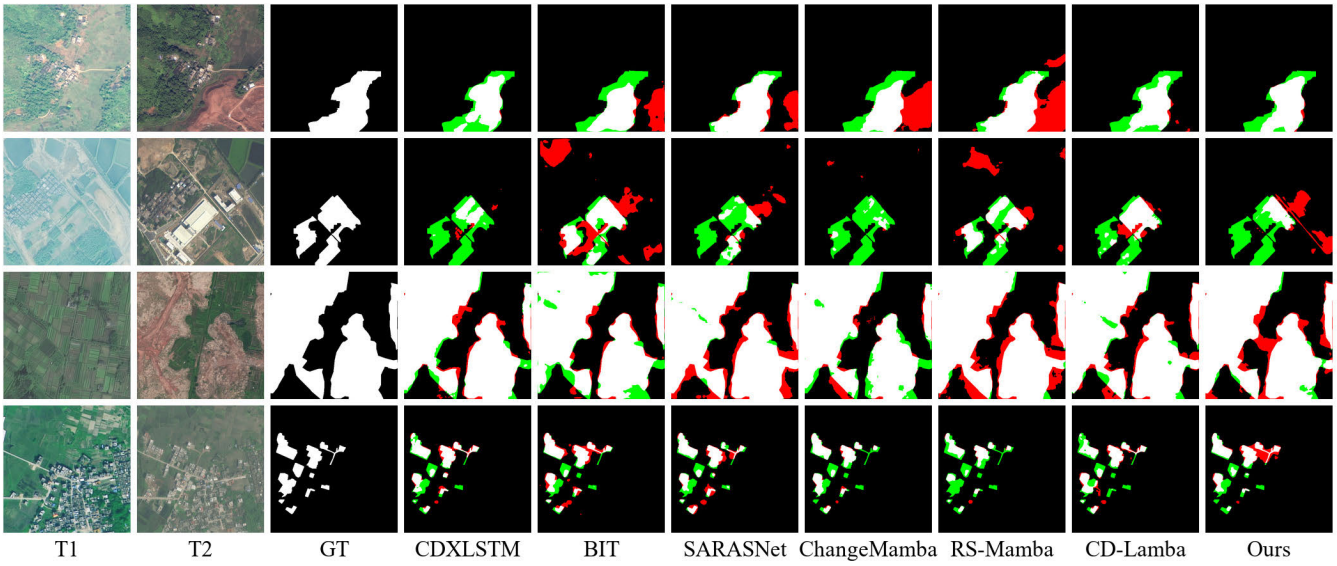


Fig. 6. Visualization results of different methods on the CLCD test set. TPs are shown in white, TNs in black, FPs in red, and FNs in green.

detection. In contrast, removing STPM or GLCD results in relatively milder degradation, indicating that these modules mainly refine and complement the learned representations.

To further analyze the internal structure of each block, key subblocks are also ablated individually. Removing MFRM from FDEM causes a substantial performance drop ($F1$ = 85.72% and OA = 94.63%), highlighting its important role in frequency-domain feature enhancement. For STPM, the performance gap between removing the channel mixing subblock and removing the entire STPM is relatively small, indicating that channel mixing mainly serves as a supportive interaction component rather than a dominant contributor. Within GLCD, removing global perception block (GPB) or local perception block (LPB) results in larger performance degradation than removing AFB, demonstrating that the collaboration between global and local perception is more critical than adaptive fusion alone.

2) Controlled Comparison of Different FDEM Variants: To further investigate whether the performance gain of FDEM arises from explicit frequency-domain modeling rather than merely from additional trainable parameters, we conduct controlled comparisons with several alternative variants on the CLCD dataset. Specifically, the proposed FDEM is compared with a fixed high-pass filtering variant, a fixed wavelet-based variant, a parameter-matched spatial enhancement branch, and the version without FDEM. As shown in Table V, all frequency-aware variants outperform the version without FDEM, indicating that explicit frequency-domain enhancement is beneficial for suppressing pseudo-changes on CLCD. Meanwhile, the proposed FDEM achieves the best overall performance among all compared variants. This result suggests that the advantage of FDEM is not merely due to increased model capacity, but is mainly attributable to its adaptive frequency-domain discrepancy modeling.

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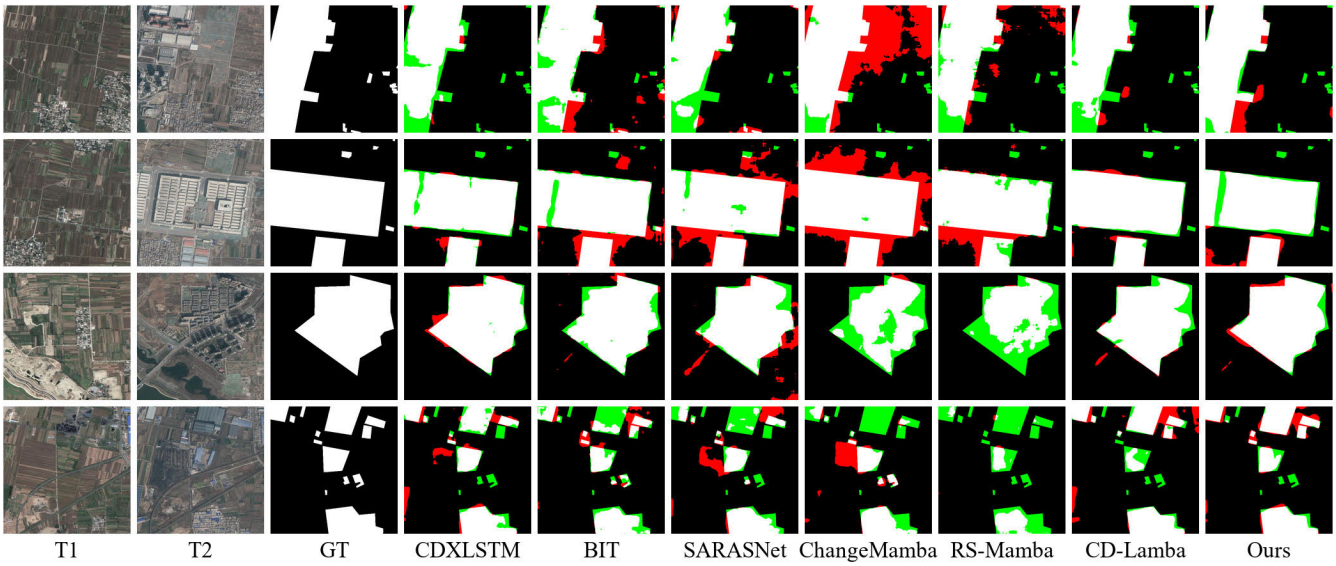


Fig. 7. Visualization results of different methods on the DSIFN-CD test set. TPs are shown in white, TNs in black, FPs in red, and FNs in green.

TABLE III
COMPREHENSIVE EFFICIENCY COMPARISON OF DIFFERENT METHODS ACROSS VARIOUS IMAGE RESOLUTIONS

Method	Resolution	Params (M)	GFLOPs	Latency (ms)	FPS	GPU Memory (MB)
SNUNet [19]	256×256	12.03	46.70	7.89	126.79	200.71
	512×512	12.03	186.79	36.95	27.06	661.96
ChangeFormer [30]	256×256	41.03	138.77	20.17	49.58	461.91
	512×512	41.03	560.02	41.25	24.24	1345.28
BiFA [31]	256×256	5.58	53.00	12.73	78.55	215.50
	512×512	5.58	212.00	31.50	31.75	685.20
CD-Lamba [58]	256×256	28.74	20.60	86.14	11.61	299.27
	512×512	28.74	82.34	119.87	8.34	691.34
FSMamba (Ours)	256×256	33.90	14.03	51.79	19.13	332.65
	512×512	33.90	45.74	57.42	17.42	549.64

TABLE IV
EFFECTS OF DIFFERENT BLOCKS ON CLCD DATASET

Model	CLCD				
	F1	Pre.	Rec.	IoU	OA
w/o FDEM	86.90	88.73	85.27	78.96	95.81
w/o MFRM	85.72	87.51	84.10	77.54	94.63
w/o STPM	87.03	89.05	85.24	78.77	96.60
w/o Channel Mixing	86.89	88.89	85.12	78.42	96.47
w/o GLCD	87.16	88.74	86.38	79.53	96.65
w/o GPB	86.05	87.62	85.21	78.18	95.58
w/o LPB	86.01	87.57	85.15	78.04	95.52
w/o AFB	86.21	88.02	85.39	78.65	95.66
Ours	87.74	89.11	86.48	79.73	96.74

3) Class Activation Mapping (CAM) Visualization of Multiscale Layers: CAM results on LEVIR-CD at activations become more change-focused and spatially coherent at deeper stages, indicating improved background suppression. See Supplementary Fig. S1 for details.

4) Effects of Different Depths in GLCD: We evaluate different decoder depths on CLCD and adopt a uniform depth of (4, 4, 4, 4), which provides the best OA–efficiency tradeoff. Detailed comparisons are reported in Supplementary Table S1.

5) Effects of Different Fusion Methods in FDEM and GLCD: We compare Add/Concat fusion strategies on CLCD and adopt Concat fusion in both FDEM and GLCD, which yields the best overall performance. Detailed results are provided in Supplementary Table S2.

6) Effects of Different Coefficients of Loss Functions: We vary the weights of Focal and Dice losses on CLCD and use an equal-weight setting ($\lambda_1 = 0.5$ and $\lambda_2 = 0.5$), which performs best overall. Detailed results are reported in Supplementary Table S3.

7) Feature Visualization Analysis of the Proposed Modules: To provide an intuitive understanding of how each proposed module contributes to the final performance, we visualize the feature maps before and after FDEM, STPM, and GLCD on representative samples from LEVIR-CD. In all visualizations, "before" and "after" denote the feature representations before and after each module, respectively. As shown in Fig. 8, the post-FDEM features exhibit more concentrated responses on changed regions with reduced scattered activations, indicating improved suppression of pseudo-changes. The post-STPM features become more continuous and less fragmented, suggesting enhanced semantic consistency between bitemporal representations. After GLCD, the decoded features present clearer building boundaries and more complete structures, demonstrating the effectiveness of the proposed global–local decoding strategy. Overall, these observations are consistent with the functional roles of the three proposed modules.

F. Generalization of FDEM Across Different Architectures

To further evaluate the transferability of the proposed FDEM across different RSCD architectures, we conduct additional experiments on the CLCD dataset from three perspectives. We first assess its effect on a strong Transformer-based baseline, then examine its applicability to representative existing Mamba-based baselines, and finally analyze whether the performance of FSMamba is sensitive to backbone choice.

1) Effect of Introducing FDEM Into a Non-Mamba Baseline: To examine whether frequency-domain enhancement is also beneficial beyond Mamba-based architectures, FDEM is inserted after each encoder stage of BiFA to enhance spectral–structural discrepancy modeling before its multilevel interaction modules. The resulting variant is denoted as BiFA + FDEM.

As shown in Table VI, introducing FDEM into BiFA consistently improves all evaluation metrics on the CLCD dataset,

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TABLE V

CONTROLLED COMPARISON OF DIFFERENT FDEM VARIANTS ON THE CLCD DATASET

Variant	Learnable	Domain	Params(M)	GFLOPs	F1	Pre.	Rec.	IoU	OA
w/o FDEM	No	Spatial	20.81	36.04	86.90	88.73	85.27	78.96	95.81
Fixed High-Pass	No	Frequency	20.81	36.04	87.42	88.74	85.42	79.45	96.67
Fixed Wavelet	No	Frequency	20.81	31.03	83.37	83.02	83.73	74.04	95.37
Param-Matched Spatial Branch	Yes	Spatial	33.87	45.72	86.56	87.66	85.23	78.13	95.40
Learnable FDEM(Ours)	Yes	Frequency	33.90	45.74	87.74	89.11	86.48	79.73	96.74

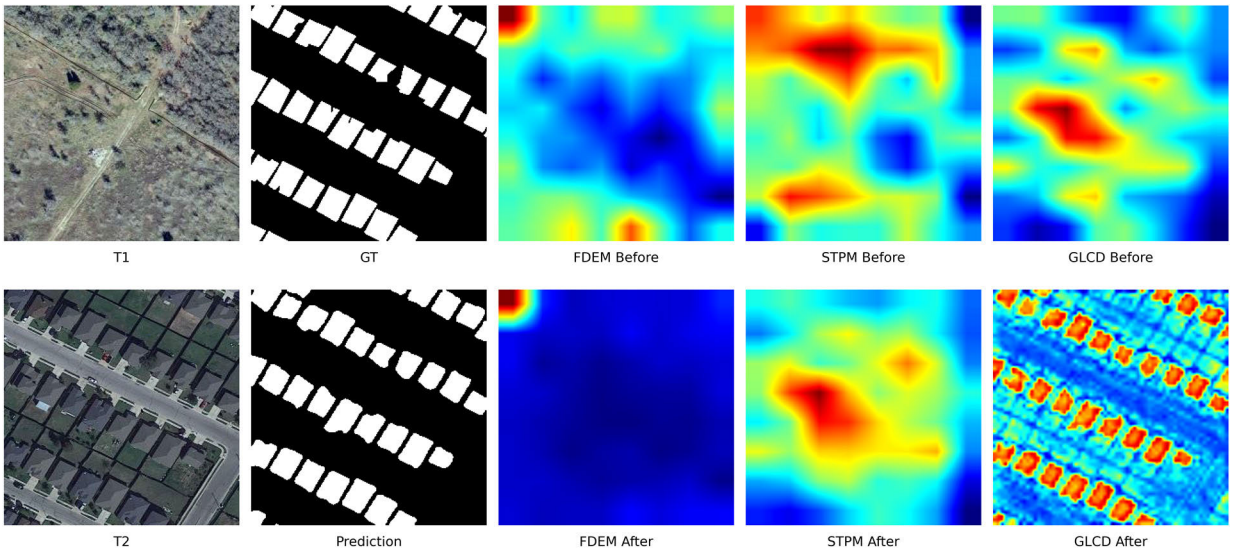


Fig. 8. Before-and-after feature visualization of the proposed FDEM, STPM, and GLCD on LEVIR-CD.

TABLE VI

EFFECTS OF INTEGRATING FDEM INTO THE TRANSFORMER BASELINE BIFA ON CLCD

Model	F1	Precision	Recall	IoU	OA
BiFA [31]	84.02	84.02	74.98	65.61	97.08
BiFA+FDEM	85.85	86.47	82.35	68.52	97.46
FSMamba (ours)	87.74	89.11	86.48	79.73	96.74

TABLE VII

EFFECT OF INSERTING FDEM INTO EXISTING MAMBA-BASED BASELINESON CLCD

Model	F1	Pre.	Rec.	IoU	OA
ChangeMamba [44]	70.00	75.39	65.33	53.85	95.83
ChangeMamba+FDEM	75.86	80.11	72.03	60.17	96.22
CDMamba [45]	71.58	79.48	69.05	56.02	95.96
CDMamba+FDEM	78.01	82.33	74.12	62.35	96.41

yielding gains of 1.83% in $F1$-score and 2.91% in IoU. This confirms that frequency-domain priors provide complementary structural cues even for a strong Transformer backbone. Nevertheless, FSMamba achieves substantially higher overall performance ($F1$ = 87.74% and IoU = 79.73%) compared with both BiFA and BiFA + FDEM. This indicates that frequency enhancement can be more effectively exploited when coupled with the SSM mechanism and the proposed global–local collaborative decoding strategy. The synergy between frequency-aware representations and input-dependent state transitions enables more stable suppression of pseudo-changes and improved structural consistency in complex RSCD scenarios.

It is worth noting that, although FSMamba achieves a higher $F1$-score and IoU than both BiFA and BiFA + FDEM on CLCD, its OA is slightly lower. This discrepancy is mainly caused by the severe class imbalance in CLCD, where unchanged pixels dominate the image. In such cases, OA is more sensitive to background classification accuracy, whereas $F1$-score and IoU more directly reflect the detection quality on changed regions. To provide a clearer interpretation, we further report classwise results and normalized confusion matrices in Supplementary Tables S4 and S5.

2) Effect of Introducing FDEM Into Existing Mamba-Based Baselines: To further evaluate whether the proposed FDEM can serve as a transferable enhancement module for existing Mamba-based RSCD methods, we additionally integrate it into ChangeMamba and CDMamba under the same experimental setting on CLCD.

As shown in Table VII, introducing FDEM consistently improves both ChangeMamba and CDMamba across all metrics. Specifically, $F1$-score increases from 70.00% to 75.86% for ChangeMamba and from 71.58% to 78.01% for CDMamba, while IoU improves from 53.85% to 60.17% and from 56.02% to 62.35%, respectively. These results further demonstrate that the proposed FDEM is not restricted to the specific architecture of FSMamba, but can be generalized as a complementary frequency-enhancement component for existing Mamba-based RSCD frameworks.

3) Effect of Different Backbones in FSMamba: To further investigate whether the performance gain of FSMamba mainly depends on the adopted backbone, we replace the SeaFormer encoder with ResNet18 and ResNet50 and conduct controlled comparisons on the CLCD dataset under the same experimental setting.

As shown in Table VIII, FSMamba remains effective under different backbone choices. Although SeaFormer yields the best overall performance, the ResNet18- and ResNet50-based variants still achieve competitive results on CLCD. This indicates that the proposed framework is not tightly coupled

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TABLE VIII
EFFECT OF DIFFERENT BACKBONES IN FSMAMBA ON THE CLCD DATASET

Model	F1	Pre.	Rec.	IoU	OA
FSMamba (SeaFormer)	87.74	89.11	86.48	79.73	96.74
FSMamba (ResNet18)	83.46	85.22	81.79	72.81	96.12
FSMamba (ResNet50)	85.19	86.73	83.72	75.43	96.35

to a specific encoder, and its gain mainly stems from the collaborative modeling of frequency-domain enhancement, spatiotemporal perception, and global–local decoding.

V. CONCLUSION

In this article, we proposed FSMamba, a frequencyenhanced spatiotemporal Mamba network for local-global remote sensing image change detection. FSMamba integrates frequency-domain enhancement, spatiotemporal state-space modeling, and global-local collaborative decoding into a unified encoder-decoder framework. The proposed FDEM suppresses pseudo-changes caused by illumination variations, noise, and misregistration errors by modeling structural discrepancies in the frequency domain. The STPM enhances long-range dependency modeling and semantic consistency between bitemporal features through Mamba-based visual state-space modeling. The GLCD further improves change-region reconstruction by adaptively integrating global contextual information and local structural details. Extensive experiments on five public benchmark datasets demonstrate that FSMamba achieves competitive or superior performance compared with state-of-the-art CNN-, Transformer-, and Mamba-based change detection methods. Ablation studies verify the effectiveness of the proposed FDEM, STPM, and GLCD, and generalization experiments further show that FDEM can serve as a transferable frequency-enhancement component for different architectures. These results confirm the effectiveness and robustness of FSMamba in complex remote sensing image change detection scenarios. In future work, we will further investigate lightweight model designs and extend the proposed framework to multimodal and multitemporal change detection tasks.

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